Jan 14

What's Computer Architecture?

Efficient design of computers

Fast circuits → VLSI nanoelectronics
→ Efficient programs → algorithms, SWE

Speed up computers using:

1. Parallelism
2. Speculation (doing things before needed)
   - 2a. with prediction
   - 2b. without prediction

1. Parallelism
    1. Gate, bit level
    2. Instruction level
    3. Thread level (programs must change)
    4. Inter-program level

n-core processor ≠ n times faster than a 1-core proc.

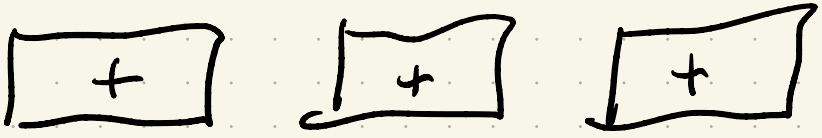
Instruction level

A

R1 ← R2+R3
R4 ← R1+R5
R6 ← R4+R7

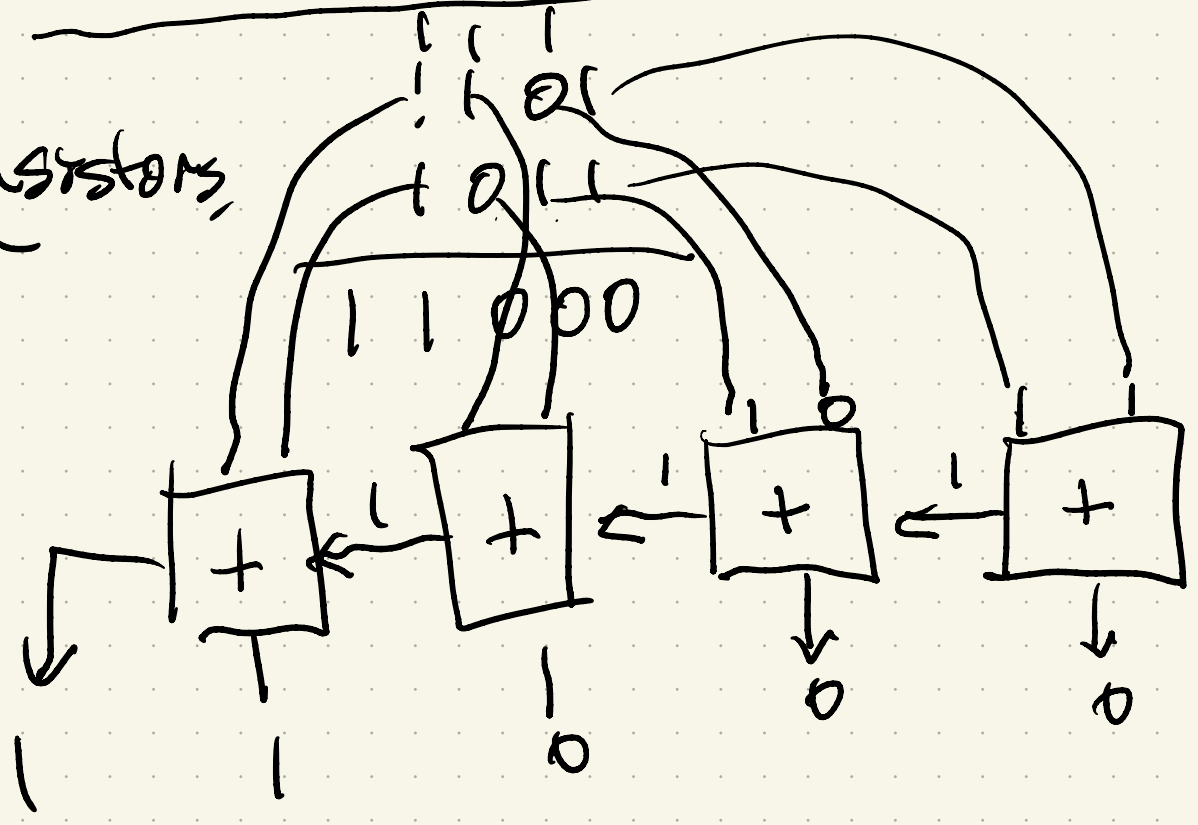
B

R1 ← R2+R3
R4 ← R6+R5
R7 ← R8+R9

computer 

Gate level

less transistors, slower



(SOP)

→ Any n-input function can be implemented in 2 levels of logic

more transistors, faster

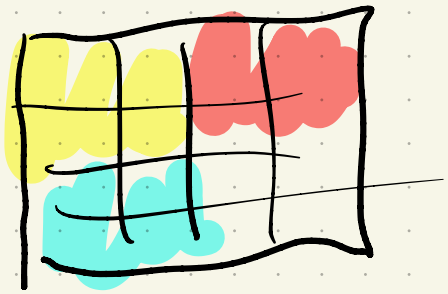
F(1,1,0,1,1,0,1,1) → 11000

Thread level parallelism

Programmer model vs. Hardware

Flynn Taxonomy

{single / motiple} Instruction streams, {Single / Multiple} data streams



SISD, MIMD

SIMD, MISD